

Personal Library & AI Research Engine — Full Capability Mindmap

Full-Stack Architectural Decomposition: Ingestion, BibTeX, Dual AI, Qdrant RAG, UI5, Security, and Governance

PERSONAL LIBRARY & AI RESEARCH ENGINE — COMPLETE FUNCTIONALITY MINDMAP

100% Comprehensive Capability Matrix: Spring Boot 3 • Spring AI • UI5 Web Components • Qdrant Vector RAG • Camunda BPMN • Keycloak OIDC

AGPL-3.0-or-later • Enterprise



Functional Capability Matrix & Technical Specifications

Detailed Pillar-by-Pillar Breakdown of the 8 Core Architectural Subsystems

1. Ingestion & Document Pipeline

Multi-format document processing with automated orchestration.

- Binary formats: PDF, DOCX, Markdown, LaTeX, Plain Text with MIME validation.
- Camunda BPMN 2.0 7-step orchestrated pipeline with human review loop.
- SHA-256 fingerprinting for atomic de-duplication and integrity verification.
- Physical versioned file system with secure streaming endpoint delivery.

2. BibTeX & LaTeX Metadata Engine

AST-based academic citation parsing and manipulation.

- Full AST tokenization of @article, @book, @inproceedings, @techreport, @misc.
- 14 structured metadata attributes: title, author, year, month, DOI, URL, abstract, etc.
- LaTeX accent sanitization (converting \'{a}, \'{e} into UTF-8 standard characters).
- Live interactive browser BibTeX editor with schema validation and clipboard export.

3. Dual AI Models & Benchmarking

Spring AI multi-model comparative synthesis engine.

- Llama 3.3 (70B Instruct): Deep academic synthesis, critique, and methodology breakdown.
- Mistral Large (2411): High-speed executive summary and core operational takeaways.
- Google Gemini API cloud integration for multimodal processing fallback.
- Synchronized side-by-side comparator UI with latency and duration telemetry.

4. Vector RAG & Conversational Chat

Sub-10ms neural search and grounded conversational Q&A.

- Semantic chunking: 500-token chunks with 50-token contextual overlap windows.
- Qdrant vector collection library_embeddings with Cosine distance metric & HNSW.
- Conversational assistant with multi-turn context memory and prompt guardrails.
- Interactive citation drawer with exact quote highlights and confidence scores.

5. Version Control & Lineage Tracking

Non-destructive in-place versioning and audit preservation.

- Multipart version overwriting that preserves the root document GUID.
- Version lineage pointer chain (versionNumber, previousVersionGuid, rootGuid).
- Complete mutation audit history: timestamps, editing user, and modified fields.
- Historical document rollback and raw binary download for any prior version.

6. Enterprise UI5 Frontend Experience

SAP Fiori Horizon floorplans built on web components.

- Official @ui5/webcomponents v2 wrapping with zero external framework dependencies.
- Analytical ListReport with multi-field FilterBar and adaptive ObjectPage with KPI header.
- Dual theme support: SAP Horizon (Light) and SAP Horizon Dark.
- Internationalization (i18n): 5 Locales (English, German, French, Spanish, Romanian).

7. Security, Identity & Governance

Enterprise OAuth2/OIDC and role-based access control.

- Keycloak 24+ integration with JWT Bearer token authentication.
- Role-Based Access Control (RBAC): LIBRARY_ADMIN, CHIEF_RESEARCHER, VIEWER.
- Security audit logging for administrative mutations and file streaming operations.
- GNU AGPL-3.0-or-later licensing protecting free and open research collaboration.

8. Architecture, API & Tooling Ecosystem

Standards-compliant documentation and exporter tooling.

- OpenAPI 3.1 REST API specification and interactive Swagger schema viewer.
- Complete C4 Architecture diagrams and Java / TypeScript UML class diagrams.
- Automated multi-format exporter: SVG, Ultra-HD PNG, Archival PDF, Draw.io, Mermaid.
- Self-contained repository ZIP exporter with full source code, scripts, and models.